Rockchip Gpio Output Clocks

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前言

概述

产品版本

芯片名称	内核版本
RK3399	linux4.4及以上
RK1808	linux4.4及以上
RK3328	linux4.4及以上
RK3308	linux4.4及以上
RV1126	linux4.19及以上
PX30	linux4.4及以上
RK3566/8	linux4.19及以上
RK3588	linux5.10
RK3528	linux4.19
RK3576	linux6.1
RK2118	RTT
RV1126B/RV1126B-P	linux6.1
RK2116	RTT
RK1820	RTT
RK3538/9	linux6.1

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

版本号	作者	修改日期	修改说明
V1.0.0	张晴	2021-09-06	第一次版本发布
V1.1.0	张晴	2021-12-29	增加RK3588支持
V1.2.0	张晴	2023-02-27	增加RK3528支持
V1.3.0	张晴	2024-04-27	增加RK3576&RK2118支持
V1.4.0	张晴	2025-04-16	增加RV1126B/RV1126B-P支持
V1.5.0	张晴	2025-07-18	增加RK2116/RK1820支持
V1.6.0	张晴	2026-01-09	增加RK3538/9支持

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1. 概念介绍

1.1 概念解释

随着外围设备增加以及成本约束，很多外设会希望使用SOC的IO输出某些特定时钟频率来替换外部晶振，从而减少外围电路的成本。

1.2 主要配置

SOC上IO输出CLK，前提是这个IO的IOMUX有CLK输出功能，

1. IOMUX

这个IO的IOMUX必须配置成CLK输出功能。

2. CLK频率

需要配置CLK频率。因为是替代外部晶振的，所有这个CLK如果是输出24M并且是从SOC的晶振bypass的，那么信号质量最好。如果是从PLL分频下来的信号质量稍微差一些，是否可以满足外设需求要看外设的参考文档。

2. IO驱动能力

有的IO上测试到的波形并不好看，可能还需要调整IO的驱动能力。

2. 对应关系

2.1 GPIO和CLK硬件对应关系

芯片名称	**clk name**	**gpio name**	支持频率
RK3288	SCLK_TESTOUT	GPIO0_C1	24M、32K
RK3328	SCLK_TESTOUT	GPIO1_A6	24M、32K
PX30	SCLK_TEST_OUT	GPIO1_D6\GPIO0_B5	24M、32K
RK3308	SCLK_TESTOUT	GPIO0_A4	24M、32K
RV1108	SCLK_TESTOUT	GPIO0_A6	24M、32K
RV1126	SCLK_TESTOUT	GPIO1_A4	24M、32K
RK3368	SCLK_TESTOUT	GPIO0_B0	24M、32K
RK3399	SCLK_TESTOUT	GPIO2_D1\GPIO0_B0	24M、32K
RK3566/8	CLK_WIFI	GPIO0_A0	24M
RK3566/8	CLK_MAC0_OUT	GPIO2_C1	24M、25M、50M、125M
RK3566/8	CLK_MAC1_OUT	GPIO3_B0	24M、25M、50M、125M
RK3566/8	CLK_MAC1_OUT	GPIO4_B3	24M、25M、50M、125M
RK3566/8	CLK_CIF_OUT	GPIO4_C0	24M、27M、37.125M
RK3566/8	CLK_CAM0_OUT	GPIO4_A7	24M、27M、37.125M
RK3588	REFCLKOUT	GPIO0_A0	24M
RK3588	CLK_GMAC_50M	GPIO4_C3	50M、125M
RK3588	REFCLK25M_ETH0	GPIO4_B0	25M、50M
RK3588	CLK_DEEPSLOW	GPIO2_C5	32K
RK3588	CLK_MIPI_CAMARAOUT_M0	GPIO4_B1	24M、27M、37.125M
RK3588	CLK_MIPI_CAMARAOUT_M1	GPIO1_B6	24M、27M、37.125M
RK3588	CLK_MIPI_CAMARAOUT_M2	GPIO1_B7	24M、27M、37.125M
RK3588	CLK_MIPI_CAMARAOUT_M3	GPIO1_D6	24M、27M、37.125M
RK3588	CLK_MIPI_CAMARAOUT_M4	GPIO1_D7	24M、27M、37.125M
RK3588	CLK_CIF_OUT	GPIO4_B4	24M、27M、37.125M
RK3588	REFCLK25M_ETH1	GPIO3_A6	25M、50M
RK3528	CLK_REFOUT	GPIO0_A1	24M

芯片名称	**clk name**	**gpio name**	支持频率
RK3528	CLK_REFOUT	GPIO3_C3	24M
RK3528	CLK_DEEPSLOW	GPIO3_C3	32K
RK3528	CLK_DEEPSLOW	GPIO1_C3	32K
RK3528	CLK_TESTOUT	GPIO2_A5	要看实际配置
RK3576	REF_CLK0_OUT	GPIO0_A0	12M、24M、27M、37.125M
RK3576	REF_CLK1_OUT	GPIO0_B4	12M、24M、27M、37.125M
RK3576	REF_CLK2_OUT	GPIO0_B5	12M、24M、27M、37.125M
RK3576	CLK_32K_OUT	GPIO0_A2	32K
RK3576	ETH_CLK0_25M_OUT_M0\M1	GPIO3_A4\GPIO2_D7	25M、27M、50M、125M
RK3576	ETH_CLK1_25M_OUT_M0\M1	GPIO2_D6\GPIO1_D5	25M、27M、50M、125M
RK3576	VI_CIF_CLKOUT	GPIO3_A2	24M、27M、37.125M
RK3576	CAM_CLK0_OUT_M0\M1	GPIO3_D7\GPIO2_D2	24M、25M、26M、27M、37.125M、50M
RK3576	CAM_CLK1_OUT_M0\M1	GPIO4_A0\GPIO2_D6	24M、25M、26M、27M、37.125M、50M
RK3576	CAM_CLK2_OUT_M0\M1	GPIO4_A1\GPIO2_D7	24M、25M、26M、27M、37.125M、50M
RK3576	UFS_REFCLK	GPIO4_D1	24M、26M
RK3576	CLK_TESTOUT	GPIO2_A5	要看实际配置
RK2118	REF_CLK0_OUT	GPIO3_B1	24M、24.576M、40M、49.152M、50M
RK2118	REF_CLK1_OUT	GPIO0_A3	24M、24.576M、40M、49.152M、50M
RK2118	ETH_CLK_25M_OUT	GPIO0_A2	25M、50M
RK2118	OSC_CLK_OUT	GPIO0_B3	晶振bypass
RK2118	CLK_32K	GPIO0_A4	32K
RK2118	CLK_TESTOUT	GPIO3_C6	要看实际配置

芯片名称	**clk name**	**gpio name**	支持频率
RV1126B/RV1126B-P	SAI0_MCLK_M0\M1	GPIO7_A2\GPIO6_A4	12.288M、24M、24.576M、49.152M
RV1126B/RV1126B-P	SAI1_MCLK_M0\M1	GPIO1_B0\GPIO4_A3	12.288M、24M、24.576M、49.152M
RV1126B/RV1126B-P	SAI2_MCLK_M0\M1	GPIO3_B6\GPIO5_A3	12.288M、24M、24.576M、49.152M
RV1126B/RV1126B-P	CAM_CLK0_OUT	GPIO4_B1	24M、25M、26M、27M、37.125M、50M
RV1126B/RV1126B-P	CAM_CLK1_OUT	GPIO4_B0	24M、25M、26M、27M、37.125M、50M
RV1126B/RV1126B-P	CAM_CLK2_OUT	GPIO4_A1	24M、25M、26M、27M、37.125M、50M
RV1126B/RV1126B-P	CAM_CLK3_OUT	GPIO4_A0	24M、25M、26M、27M、37.125M、50M
RV1126B/RV1126B-P	VI_CIF_CLKOUT_M0\M1	GPIO6_C2\GPIO5_C5	24M、25M、26M、27M、37.125M、50M
RV1126B/RV1126B-P	ETH_CLK_25M_OUT_M0\M1	GPIO6_C1\GPIO5_C1	24M、25M、26M、27M、37.125M、50M
RV1126B/RV1126B-P	CLK_32K_OUT	GPIO0_A2	32K
RV1126B/RV1126B-P	CLK_REFOUT	GPIO0_A0	24M
RK2116	ETH_CLK_25M_OUT	GPIO0_A2	25M、50M
RK2116	OSC_CLK_OUT	GPIO0_A4	晶振bypass
RK2116	CLK_32K	GPIO0_A1	32K
RK2116	CLK_TESTOUT	GPIO2_A3	要看实际配置
RK1820	CLK_TESTOUT0	GPIO0_A0	要看实际配置
RK1820	CLK_TESTOUT1	GPIO0_A2	要看实际配置
RK3538/9	REF_CLK0_OUT	GPIO0_A1	12M、24M、27M、37.125M
RK3538/9	REF_CLK1_OUT	GPIO5_C2	12M、24M、27M、37.125M

芯片名称	clk name	gpio name	支持频率
RK3538/9	CLK_32K_OUT	GPIO3_C2\GPIO0_D4	32K
RK3538/9	ETH_CLK0_25M_OUT_M0\M1	GPIO0_D5\GPIO0_B4	25M、27M、50M、125M
RK3538/9	CLK_TESTOUT	GPIO2_A5\GPIO0_A1	要看实际配置

3. 芯片示例

3.1 IO命令示例

3.1.1 RK3288

1. Test_clk 输出24M，测试点GPIO0_C1

```
io -4 0xff760068 0x0f000000 # test clk div set 1
io -4 0xff760170 0x80000000 # enable test clk
io -4 0xff7601e8 0x0f000800 # test clk select 24m
io -4 0xff73008c 0x000c0005 # gpio0_c1 iomux select testclk
```

3.1.2 RK3328

1. Test_clk 输出24M，测试点GPIO1_A6

```
io -4 0xff440108 0x000f0000 # test clk div set 1
io -4 0xff440200 0x02000000 # enable test clk
io -4 0xff100010 0x30002000 # gpio1_a6 iomux select testclk
io -4 0xff440084 0x1f001700 # test clk select 24m
```

3.1.3 PX30

1. Test_clk 输出24M，测试点GPIO1_D6或者GPIO0_B5

```
io -4 0xff2b0240 0x80000000 # enable test clk
io -4 0xff2b01e4 0x1f1f0007 # test clk select 24m and div set 1
io -4 0xff14001c 0x0f000300 # gpio1_d6 iomux select testclk
io -4 0xff010004 0x0c000800 # gpio0_b5 iomux select testclk
```

3.1.4 RK3308

1. Test_clk 输出24M，测试点GPIO0_A4

```
io -4 0xff500224 0x1fff1700 # test clk select 24m and div set 1
io -4 0xff500310 0x02000000 # enable test clk
io -4 0xff000000 0x03000100 # gpio0_a4 iomux select testclk
```

3.1.5 RV110X

1. Test_clk 输出24M，测试点GPIO0_A6

```
io -4 0x202000fc 0x1f000000 # test clk div set 1
io -4 0x20200144 0x02000000 # enable test clk
io -4 0x202001cc 0x0f000800 # test clk select 24m
io -4 0x20060000 0x30002000 # gpio0_a6 iomux select testclk
```

3.1.6 RV1126

1. Test_clk 输出24M，测试点GPIO1_A4

```
io -4 0xff49022c 0xffff0000 # test clk select 24m and div set 1
io -4 0xff4902dc 0x20000000 # enable test clk
io -4 0xfe010014 0x00070002 # gpio1_a4 iomux select testclk
```

3.1.7 RK3368

1. Test_clk 输出24M，测试点GPIO0_B0

```
io -4 0xff7601bc 0x1f000000 # test clk div set 1
io -4 0xff76021c 0x00020000 # enable test clk
io -4 0xff760380 0x000f0008 # test clk select 24m
io -4 0xff738004 0x00030001 # gpio0_b0 iomux select testclk
```

3.1.8 RK3399

1. Test_clk 输出24M，测试点GPIO2_D1或者GPIO0_B0

```
echo 24000000 > d/clk/clk_testout1_pll_src/clk_rate
echo 24000000 > d/clk/clk_testout1/clk_rate # test clk set 24m
echo 1 > d/clk/clk_testout1/clk_enable_count # enable test clk
io -4 0xff77e004 0x000c0008 # gpio2_d1 iomux select testclk

echo 24000000 > d/clk/clk_testout2_pll_src/clk_rate
echo 24000000 > d/clk/clk_testout2/clk_rate # test clk set 24m
echo 1 > d/clk/clk_testout2/clk_enable_count # enable test clk
io -4 0xff320004 0x00030003 # gpio0_b0 iomux select testclk
```

3.1.9 RK3566/8

IO输出CLK功能更加强大

1. CLK32K_OUT0 测试点gpio0_b0

```
io -4 0xfdd00100 0x00c00080
io -4 0xfdc20100 0x00010000
io -4 0xfdc20008 0x00030002
```

2. CLK32K_OUT1 测试点gpio2_c6

```
io -4 0xfdd00100 0x00c00080
io -4 0xfdc20100 0x00010000
io -4 0xfdc20008 0x00030002
io -4 0xfdc60034 0x07000100
```

3. REF_CLKOUT 测试点gpio0_a0

```
io -4 0xfdc20000 0x00070001
echo 24000000 >/sys/kernel/debug/clk/clk_wifi/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_wifi/clk_enable_count
```

4. ETH_REFCLK_25M 测试点gpio2_c1

```
io -4 0xfdc60030 0x00700020
echo 25000000 >/sys/kernel/debug/clk/clk_mac0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac0_out/clk_enable_count
```

5. ETH_REFCLK_25M_M0 测试点gpio3_b0

```
io -4 0xfdc60048 0x00070003
echo 25000000 >/sys/kernel/debug/clk/clk_mac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac1_out/clk_enable_count
```

6. ETH_REFCLK_25_M1 测试点gpio4_b3

```
io -4 0xfdc60068 0x70003000
echo 25000000 >/sys/kernel/debug/clk/clk_mac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac1_out/clk_enable_count
```

7. CIF_CLKOUT 测试点gpio4_c0

```
io -4 0xfdc60070 0x00070001
echo 27000000 >/sys/kernel/debug/clk/clk_cif_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cif_out/clk_enable_count
```

8. CAM_CLKOUT0 测试点gpio4_a7

```
io -4 0xfdc60064 0x70001000
echo 27000000 >/sys/kernel/debug/clk/clk_cam0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cam0_out/clk_enable_count
```

9. CAM_CLKOUT1 测试点gpio4_b0

```
io -4 0xfdc60068 0x00070001
echo 24000000 >/sys/kernel/debug/clk/clk_cam1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cam1_out/clk_enable_count
```

10. TESTCLK 测试点gpio2_a2

```
io -4 0xfdc60020 0x07000200
io -4 0xfdd20228 0x1fff0000
io -4 0xfdd20388 0x80000000
```

备注：如果想测试其他频率或者频率点，按照上面寄存器查找TRM，有详细说明。

3.1.10 RK3588

IO输出CLK功能更加强大

1. GPIO0_A0 -> refclkout 24M xin_osc0_func

```
io -4 0xfd5f0000 0x000f0001
```

2. GPIO2_C3 -> eth0refclk 50/25M refclko25m_eth0

```
echo 25000000 > /sys/kernel/debug/clk/refclko25m_eth0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_eth0_out/clk_prepare_enable
io -4 0xfd5f8050 0xf00001000
```

3. GPIO4_C3 -> gmac0_clkout 125/50M clk_gmac_50m_cru_i

```
echo 50000000 > /sys/kernel/debug/clk/clk_gmac_50m/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_gmac_50m/clk_prepare_enable
echo 125000000 > /sys/kernel/debug/clk/clk_gmac_125m/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_gmac_125m/clk_prepare_enable
io -4 0xfd5f8090 0xf0001000
```

4. GPIO2_C5 -> 32kout 32K clk_deepslow

```
io -4 0xfd5f8054 0x00f00010
```

5. gpio4_b1 -> clk_mipi_camaraout_m0

```
io -4 0xfd5f8088 0x00f00010
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m0/clk_rate
```

6. gpio1_b6 -> clk_mipi_camaraout_m1

```
io -4 0xfd5f802c 0x0f000200
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m1/clk_rate
```

7. gpio1_b7 -> clk_mipi_camaraout_m2

```
io -4 0xfd5f802c 0xf0002000
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m2/clk_rate
```

8. gpio1_d6 -> clk_mipi_camaraout_m3

```
io -4 0xfd5f803c 0x0f000200
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m3/clk_rate
```

9. gpio1_d7 -> clk_mipi_camaraout_m4

```
io -4 0xfd5f803c 0xf0002000
echo 24000000 > /sys/kernel/debug/clk/clk_mipi_camaraout_m4/clk_rate
```

10. gpio4_b4 -> clk_cifout_out

```
io -4 0xfd5f808c 0x000f0001
echo 24000000 > /sys/kernel/debug/clk/clk_cifout_out/clk_rate
```

11. gpio3_a6 -> eth1refclk 50/25M refclko25m_eth0

```
echo 25000000 > /sys/kernel/debug/clk/refclko25m_eth1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_eth1_out/clk_prepare_enable
io -4 0xfd5f8064 0x0f000100
```

12. gpio4_d5 -> testclkoutm0

litcore 10分频

```
io -4 0xfd818314 0x00ff0009
io -4 0xfd7c0328 0x0e000600
io -4 0xfd818808 0x200000
io -4 0xfd7c0808 0x500000
io -4 0xfd5f809c 0x00f00040
```

3.1.11 RK3528

1. REF_CLK_OUT_M0 测试点gpio0_a1

```
io -4 0xFF4B0808 0x00100000
io -4 0xFF540000 0x00f00010
```

2. REF_CLK_OUT_M2 测试点gpio3_c3

```
io -4 0xFF4B0808 0x00100000
io -4 0xFF560070 0xf0006000
```

3. CLK_32K_OUT_M0 测试点gpio3_c3

```
io -4 0xFF560070 0xf0003000
```

4. CLK_32K_OUT_M1 测试点gpio1_c3

```
io -4 0xFF560030 0xf0001000
```

5. TESTCLK 测试点gpio2_a5

gpll 10分频输出到testclk

```
io -4 0xFF570044 0x00f00060
io -4 0xFF4A0394 0x03e00120
io -4 0xFF4A0394 0x3c000000
```

3.1.12 RK3576

1. ref_clk0_out 测试点 gpio0a0

```
echo 27000000 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_rate
echo 1 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_prepare_enable
io -4 0x26024000 0x0c000000
io -4 0x26040000 0x000f0001
```

晶振Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_prepare_enable
io -4 0x26024000 0x0c000400
io -4 0x26040000 0x000f0001
```

晶振/2 Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk0_out_pll/clk_prepare_enable
io -4 0x26024000 0x0c000800
io -4 0x26040000 0x000f0001
```

2. ref_clk1_out 测试点 gpio0b4

```
echo 24000000 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_rate
echo 1 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_prepare_enable
io -4 0x26024000 0x30000000
io -4 0x26042000 0x000f0001
```

晶振Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_prepare_enable
io -4 0x26024000 0x30001000
io -4 0x26042000 0x000f0001
```

晶振/2 Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk1_out_pll/clk_prepare_enable
io -4 0x26024000 0x30002000
io -4 0x26042000 0x000f0001
```

3. ref_clk2_out 测试点 gpio0b5

```
echo 50000000 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_rate
echo 1 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_prepare_enable
io -4 0x26024000 0xc0000000
io -4 0x26042000 0x00f00010
```

晶振Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_prepare_enable
io -4 0x26024000 0xc0004000
io -4 0x26042000 0x00f00010
```

晶振/2 Bypass

```
echo 1 > /sys/kernel/debug/clk/ref_clk2_out_pll/clk_prepare_enable
io -4 0x26024000 0xc0008000
io -4 0x26042000 0x00f00010
```

4. clk0_32k_out输出 测试点 gpio0a2

```
echo 1 > /sys/kernel/debug/clk/xin_osc0_div/clk_prepare_enable
io -4 0x26040000 0x0f000a00
io -4 0x26024000 0x00060000
```

5. ETH_CLK0_25M_OUT_M0 测试点 gpio3a4

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_prepare_enable
io -4 0x26044064 0x000f0003
```

ETH_CLK0_25M_OUT_M1 测试点 gpio2d7

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac0_out/clk_prepare_enable
io -4 0x2604405c 0xf0003000
```

6. ETH_CLK1_25M_OUT_M0 测试点 gpio2d6

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_prepare_enable
io -4 0x2604405c 0x0f000200
```

ETH_CLK1_25M_OUT_M1 测试点 gpio1d5

```
echo 50000000 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_rate
echo 1 > /sys/kernel/debug/clk/refclko25m_gmac1_out/clk_prepare_enable
io -4 0x2604403c 0x00f00010
```

7. VI_CIF_CLKOUT 测试点 gpio3a2

```
echo 27000000 > /sys/kernel/debug/clk/clk_cifout_out/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cifout_out/clk_prepare_enable
io -4 0x26044060 0x0f000100
```

8. CAM_CLK0_OUT M0 测试点 gpio3d7

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_prepare_enable
io -4 0x2604407c 0xf0003000
```

CAM_CLK0_OUT M1 测试点 gpio2d2

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m0/clk_prepare_enable
io -4 0x26044058 0x0f000100
```

9. CAM_CLK1_OUT M0 测试点 gpio4a0

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_prepare_enable
io -4 0x26044080 0x000f0003
```

CAM_CLK1_OUT M1 测试点 gpio2d6

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m1/clk_prepare_enable
io -4 0x2604405c 0x0f000100
```

10. CAM_CLK2_OUT M0 测试点 gpio4a1

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_prepare_enable
io -4 0x26044080 0x00f00030
```

CAM_CLK2_OUT M1 测试点 gpio2d7

```
echo 27000000 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi_cameraout_m2/clk_prepare_enable
io -4 0x2604405c 0xf0001000
```

11. UFS_REFCLK 测试点 gpio4d1

```
echo 26000000 > /sys/kernel/debug/clk/clk_ref_ufs_clkout/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_ref_ufs_clkout/clk_prepare_enable
io -4 0x2722030c 0x00030002
io -4 0x26004b398 0x00f00010
```

12. CLK_TEST_OUT 测试点 gpio2a5

gpll 10分频

```
io -4 0x2720036c 0x00070001
io -4 0x27200368 0xfffc0024
io -4 0x27200808 0x0f000000
io -4 0x26044044 0x00f00050
```

3.1.13 RK2118

1. ref_clk0_out gpio0_b3 1分频

晶振bypass

```
io -4 0x50b10308 0x00ff0000
io -4 0x50bc0a00 0x00030002
```

GPLL 出40M

```
io -4 0x50b10308 0x00ff0053
io -4 0x50bc0a00 0x00030002
```

V0PLL 出49.152M

```
io -4 0x50b10308 0x00ff0097
io -4 0x50bc0a00 0x00030002
```

V1PLL 出45.1584M

```
io -4 0x50b10308 0x00ff00d3
io -4 0x50bc0a00 0x00030002
```

2. ref_clk1_out gpio0_a3 1分频

晶振bypass

```
io -4 0x50b10308 0xff000000
io -4 0x50bc0000 0xf0001000
```

GPLL 出40M

```
io -4 0x50b10308 0xff005300
io -4 0x50bc0000 0xf0001000
```

V0PLL 出49.152M

```
io -4 0x50b10308 0xff009700
io -4 0x50bc0000 0xf0001000
```

V1PLL 出45.1584M

```
io -4 0x50b10308 0xff00d300
io -4 0x50bc0000 0xf0001000
```

3. ETH_CLK_25M_OUT gpio0_a2

```
io -4 0x50b10304 0xfc007c00
io -4 0x50bc0000 0x0f000100
```

4. OSC_CLK_OUT gpio0_b3 晶振bypass

```
io -4 0x50bc0a00 0x00030001
```

5. CLK_32K gpio0_a4

```
io -4 0x50B30000 0x00040004
io -4 0x50bc0004 0x000f0001
```

6. CLK_TESTOUT gpio3_c6

xin24 1分频

```
io -4 0x50b0030c 0x01ff0000
io -4 0x50b00800 0x10000000
io -4 0x501d0074 0x0f000300
```

gpll 10分频

```
io -4 0x50b0030c 0x01ff0029
io -4 0x50b00800 0x10000000
io -4 0x501d0074 0x0f000300
```

3.1.14 RV1126B/RV1126B-P

1. SAI0_MCLK_M0 gpio7_a2

```
io -4 0x201e00e0 0x0f000100
echo 24576000 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_prepare_enable
```

2. SAI0_MCLK_M1 gpio6_a4

```
io -4 0x201d80c4 0x000f0004
echo 24576000 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai0_out2io/clk_prepare_enable
```

3. SAI1_MCLK_M0 gpio1_b0

```
io -4 0x201b0028 0x000f0002
echo 49152000 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_prepare_enable
```

4. SAI1_MCLK_M1 gpio4_a3

```
io -4 0x201c8080 0xf0002000
echo 49152000 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_prepare_enable
```

5. SAI2_MCLK_M0 gpio3_b6

```
io -4 0x201c006c 0x0f000200
echo 12288000 > /sys/kernel/debug/clk/mclk_sai2_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai2_out2io/clk_prepare_enable
```

6. SAI2_MCLK_M1 gpio5_a3

```
io -4 0x201d00a0 0xf0005000
echo 12288000 > /sys/kernel/debug/clk/mclk_sai1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/mclk_sai2_out2io/clk_prepare_enable
```

7. CAM_CLK0_OUT gpio4_b1

```
io -4 0x201c8088 0x00f00030
echo 24000000 > /sys/kernel/debug/clk/clk_mipi0_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi0_out2io/clk_prepare_enable
```

8. CAM_CLK1_OUT gpio4_b0

```
io -4 0x201c8088 0x000f0003
echo 24000000 > /sys/kernel/debug/clk/clk_mipi1_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi1_out2io/clk_prepare_enable
```

9. CAM_CLK2_OUT gpio4_a1

```
io -4 0x201c8080 0x00f00030
echo 27000000 > /sys/kernel/debug/clk/clk_mipi2_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi2_out2io/clk_prepare_enable
```

10. CAM_CLK3_OUT gpio4_a0

```
io -4 0x201c8080 0x000f0003
echo 27000000 > /sys/kernel/debug/clk/clk_mipi3_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mipi3_out2io/clk_prepare_enable
```

11. VI_CIF_CLKOUT_M0 gpio6_c2

```
io -4 0x201d80d0 0x0f000100
echo 27000000 > /sys/kernel/debug/clk/clk_cif_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cif_out2io/clk_prepare_enable
```

12. VI_CIF_CLKOUT_M1 gpio5_c5

```
io -4 0x201d00b4 0x00f00030
echo 27000000 > /sys/kernel/debug/clk/clk_cif_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_cif_out2io/clk_prepare_enable
```

13. ETH_CLK_25M_OUT_M0 gpio6_c1

```
io -4 0x201d80d0 0x00f00020
echo 25000000 > /sys/kernel/debug/clk/clk_mac_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac_out2io/clk_prepare_enable
```

14. ETH_CLK_25M_OUT_M1 gpio5_c1

```
io -4 0x201d00b0 0x00f00020
echo 25000000 > /sys/kernel/debug/clk/clk_mac_out2io/clk_rate
echo 1 > /sys/kernel/debug/clk/clk_mac_out2io/clk_prepare_enable
```

15. clk_32k_io 1分频 test_clk_out gpio0_a2

```
io -4 0x20040300 0xfff03000
io -4 0x20040800 0x80000000
io -4 0x20040300 0x000f0000
io -4 0x20040800 0x40000000
io -4 0x201a0000 0x0f000200
io -4 0x20130004 0x01000000
```

16. clk_refout 1分频 test_clk_out gpio0_a0

```
io -4 0x20040300 0xfff00800
io -4 0x20040800 0x80000000
io -4 0x20040300 0x000f0000
io -4 0x20040800 0x40000000
io -4 0x201a0000 0x000f0002
echo 1 > /sys/kernel/debug/clk/clk_refout/clk_prepare_enable
```

3.1.15 RK2116

1. ref_clk0_out gpio0_a4 1分频

晶振bypass

```
io -4 0x40810004 0x000f0001
```

2. ETH_CLK_25M_OUT gpio0_a2

```
io -4 0x40810000 0x0f000100
```

3. CLK_32K gpio0_a1

```
io -4 0x40810000 0x00f00010
io -4 0x40800000 0x00040004
```

4. CLK_TESTOUT gpio2_a3

xin24 1分频

```
io -4 0x40870310 0x01ff0000
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

gpll 10分频

```
io -4 0x40870310 0x01ff0029
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

v0pll 10分频

```
io -4 0x40870310 0x01ff0049
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

v1pll 10分频

```
io -4 0x40870310 0x01ff0069
io -4 0x40870800 0x20000000
io -4 0x40510040 0xf0004000
```

3.1.16 RK1820

1. CLK_TESTOUT0 gpio0_a0

gpll 10分频

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80048
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x00070000
io -4 0x1030400000 0x000f0002
```

mpll 10分频

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80248
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x00070000
io -4 0x1030400000 0x000f0002
```

CLK_32K 1分频

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80800
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x00070000
io -4 0x1030400000 0x000f0002
```

2. CLK_TESTOUT1 gpio0_a2

dpll 10分频

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80448
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x1c000000
io -4 0x1030400000 0x0f000200
```

l0pll 10分频

```
io -4 0x1030800810 0xffff0000
io -4 0x1030800364 0x7ff80648
io -4 0x103080036c 0x00e00000
io -4 0x1030800368 0x1c000000
io -4 0x1030400000 0x0f000200
```

3.1.17 RK3538/9

1. REF_CLK_OUT_M0 测试点gpio0_a1

```
io -4 0xfd1b0000 0x00f00010
echo 24000000 > /sys/kernel/debug/clk/clk_refout/clk_rate
echo 1 >/sys/kernel/debug/clk/clk_refout/clk_prepare_enable
```

2. REF_CLK_OUT_M1 测试点gpio5_c2

```
io -4 0xfd2100b0 0x0f000200
echo 49500000 > /sys/kernel/debug/clk/clk_refout/clk_rate
echo 1 >/sys/kernel/debug/clk/clk_refout/clk_prepare_enable
```

3. ETH0_CLK_25M_OUT_M0 测试点gpio0_d5

```
io -4 0xfd1c001c 0x00f00010
echo 25000000 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_rate
echo 1 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_prepare_enable
```

4. ETH0_CLK_25M_OUT_M1 测试点gpio0_b4

```
io -4 0xfd1b000c 0x000f0001
echo 50000000 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_rate
echo 1 >/sys/kernel/debug/clk/eth0_clk_25m_out/clk_prepare_enable
```

5. CLK_32K_OUT_M0 测试点gpio3_c2

```
io -4 0xfd1f0070 0x0f000300
```

6. CLK_32K_OUT_M1 测试点gpio5_c2

```
io -4 0xfd2100b0 0x0f000600
```

7. CLK_32K_OUT_M2 测试点gpio0_d4

```
io -4 0xfd1c001c 0x000f0001
```

8. CLK_TEST_OUT_M0 测试点gpio0_a1

sclk_uart1 1分频

```
io -4 0xfd070308 0xf000b000
io -4 0xfd070808 0x00400000
io -4 0xfd040300 0x0fff0003
io -4 0xfd04080c 0x00070000
io -4 0xfd1b0000 0x00f00020
```

9. CLK_TEST_OUT_M1 测试点gpio2_a5

mclk_sai2 1分频

```
io -4 0xfd070308 0xf0006000
io -4 0xfd070808 0x00400000
io -4 0xfd060304 0x03ff0100
io -4 0xfd060808 0x03100000
io -4 0xfd1e0044 0x00f00020
```

备注：如果想测试其他频率或者频率点，按照上面寄存器查找TRM，有详细说明。

3.2 软件示例

先确认IO是否支持CLK输出（IOMUX功能可以看到），确认CLK支持哪些频率。如果是TESTCLK功能的，需要查询寄存器，选择不同的功能输出。如果是特定功能的，直接设置频率就可以。

3.2.1 RK3288

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT_SRC>;
assigned-clock-parents = <&xin24m>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL：

```
diff --git a/drivers/clk/rockchip/clk-rk3288.c b/drivers/clk/rockchip/clk-
rk3288.c
index 2784a7ed05db..a45afa6897a7 100644
--- a/drivers/clk/rockchip/clk-rk3288.c
+++ b/drivers/clk/rockchip/clk-rk3288.c
@@ -204,6 +204,12 @@ PNAME(mux_hsadcout_p)       = { "hsadc_src", "ext_hsadc" };
 PNAME(mux_edp_24m_p)    = { "ext_edp_24m", "xin24m" };
```

```
 PNAME(mux_tspout_p)     = { "cpll", "gpll", "npll", "xin27m" };

+PNAME(mux_testout_src_p) = { "aclk_peri", "armclk", "aclk_vio0", "ddrphy",
+                             "aclk_vcodec", "aclk_gpu", "sclk_rga", "aclk_cpu",
+                             "xin24m", "xin27m", "xin32k", "clk_wifi",
+                             "dclk_vop0", "dclk_vop1", "sclk_isp_jpe",
+                             "sclk_isp" };
+
 PNAME(mux_usbphy480m_p)                 = { "sclk_otgphy1_480m",
"sclk_otgphy2_480m",
                                     "sclk_otgphy0_480m" };
 PNAME(mux_hsicphy480m_p)        = { "cpll", "gpll", "usbphy480m_src" };
@@ -560,6 +566,12 @@ static struct rockchip_clk_branch rk3288_clk_branches[]
__initdata = {
                        RK3288_CLKSEL_CON(2), 0, 6, DFLAGS,
                        RK3288_CLKGATE_CON(2), 7, GFLAGS),

+       MUX(SCLK_TESTOUT_SRC, "sclk_testout_src", mux_testout_src_p, 0,
+           RK3288_MISC_CON, 8, 4, MFLAGS),
+       COMPOSITE_NOMUX(SCLK_TESTOUT, "sclk_testout", "sclk_testout_src", 0,
+                       RK3288_CLKSEL_CON(2), 8, 5, DFLAGS,
+                       RK3288_CLKGATE_CON(4), 15, GFLAGS),
+
        COMPOSITE_NOMUX(SCLK_SARADC, "sclk_saradc", "xin24m", 0,
                        RK3288_CLKSEL_CON(24), 8, 8, DFLAGS,
                        RK3288_CLKGATE_CON(2), 8, GFLAGS),
diff --git a/include/dt-bindings/clock/rk3288-cru.h b/include/dt-
bindings/clock/rk3288-cru.h
index 1f9c62f07389..61ae793438b4 100644
--- a/include/dt-bindings/clock/rk3288-cru.h
+++ b/include/dt-bindings/clock/rk3288-cru.h
@@ -100,6 +100,8 @@
 #define SCLK_MAC_PLL           150
 #define SCLK_MAC               151
 #define SCLK_MACREF_OUT                152
+#define SCLK_TESTOUT_SRC       153
+#define SCLK_TESTOUT           154

 #define DCLK_VOP0              190
 #define DCLK_VOP1              191
```

3.2.2 RK3328

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL：

```
diff --git a/drivers/clk/rockchip/clk-rk3328.c b/drivers/clk/rockchip/clk-
rk3328.c
index 1310cfe7798b..5b4ab3360e56 100644
--- a/drivers/clk/rockchip/clk-rk3328.c
+++ b/drivers/clk/rockchip/clk-rk3328.c
@@ -222,6 +222,14 @@ PNAME(mux_mac2io_ext_p)            = { "clk_mac2io",
                                        "gmac_clkin" };
 PNAME(mux_i2s_plls_p)          = { "cpll", "dummy_gpll" };

+PNAME(mux_sclk_testout_p) =    { "clk_wifi", "dummy", "armclk", "sclk_ddrc",
+                                 "aclk_rkvdec_pre", "aclk_rkvenc",
"aclk_vpu_pre",
+                                 "aclk_rga_pre", "aclk_vio_pre",
"aclk_vop_pre",
+                                 "aclk_gpu_pre", "aclk_bus_pre",
"aclk_peri_pre",
+                                 "aclk_gmac", "dclk_lcdc", "clk_pdm",
"clk_rga",
+                                 "sclk_vdec_core", "sclk_venc_core", "clk_tsp",
+                                 "dummy", "dummy", "dummy", "xin24m"};
+
 static struct rockchip_pll_clock rk3328_pll_clks[] __initdata = {
        [apll] = PLL(pll_rk3328, PLL_APLL, "apll", mux_pll_p,
                     0, RK3328_PLL_CON(0),
@@ -836,6 +844,10 @@ static struct rockchip_clk_branch rk3328_clk_branches[]
__initdata = {
            RK3328_SDMMC_EXT_CON0, 1),
        MMC(SCLK_SDMMC_EXT_SAMPLE, "sdmmc_ext_sample", "clk_sdmmc_ext",
            RK3328_SDMMC_EXT_CON1, 1),
+
+       COMPOSITE_DIV_OFFSET(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+               RK3328_MISC_CON, 8, 5, MFLAGS, RK3328_CLKSEL_CON(2), 0, 5,
DFLAGS,
+               RK3328_CLKGATE_CON(0), 9, GFLAGS),
 };

 static const char *const rk3328_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rk3328-cru.h b/include/dt-
bindings/clock/rk3328-cru.h
index 62479fddb96b..12f205e31273 100644
--- a/include/dt-bindings/clock/rk3328-cru.h
+++ b/include/dt-bindings/clock/rk3328-cru.h
@@ -98,6 +98,7 @@
 #define SCLK_MAC2IO            100
 #define SCLK_MAC2PHY           101
 #define SCLK_MAC2IO_EXT                102
+#define SCLK_TESTOUT           103

 /* dclk gates */
 #define DCLK_LCDC              120
```

3.2.3 PX30

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TEST_OUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TEST_OUT>;
clock-names = "soc_24M";
```

2. KERNEL：

```
diff --git a/drivers/clk/rockchip/clk-px30.c b/drivers/clk/rockchip/clk-px30.c
index 9f7a1f91627a..cc4c2fab875d 100644
--- a/drivers/clk/rockchip/clk-px30.c
+++ b/drivers/clk/rockchip/clk-px30.c
@@ -189,6 +189,13 @@ PNAME(mux_wifi_pmu_p)              = { "xin24m",
"clk_wifi_pmu_src" };
 PNAME(mux_uart0_pmu_p)          = { "clk_uart0_pmu_src", "clk_uart0_np5",
"clk_uart0_frac" };
 PNAME(mux_usbphy_ref_p)                = { "xin24m", "clk_ref24m_pmu" };
 PNAME(mux_mipidsiphy_ref_p)     = { "xin24m", "clk_ref24m_pmu" };
+PNAME(mux_sclk_test_out_p)      = { "armclk", "aclk_gpu", "clk_ddrphy4x",
"clk_i2c0", "aclk_vo_pre",
+                                    "aclk_rga", "dclk_vopb", "dclk_vopl",
"aclk_vpu_pre", "aclk_vi_pre",
+                                    "clk_isp", "clk_rtc32k_frac",
"clk_ddrphy1x", "aclk_peri_pre",
+                                    "dummy", "dummy", "dummy", "dummy",
"clk_pwm0",
+                                    "dummy", "aclk_crypto_pre",
"clk_crypto_apk", "xin24m", "aclk_gmac_pre",
+                                    "clk_gmac", "aclk_bus_pre", "clk_pdm",
"clk_i2s0_tx_out", "clk_tsadc",
+                                    "clk_uart1", "clk_saradc", "clk_otp"};

 static struct rockchip_pll_clock px30_pll_clks[] __initdata = {
        [apll] = PLL(pll_rk3328, PLL_APLL, "apll", mux_pll_p,
@@ -907,6 +914,11 @@ static struct rockchip_clk_branch px30_clk_branches[]
__initdata = {
                        PX30_CLKGATE_CON(8), 1, GFLAGS),
        GATE(PCLK_GMAC, "pclk_gmac", "pclk_gmac_pre", 0,
                        PX30_CLKGATE_CON(8), 3, GFLAGS),
+
+       COMPOSITE(SCLK_TEST_OUT, "sclk_test_out", mux_sclk_test_out_p,
CLK_SET_RATE_NO_REPARENT,
+                       PX30_CLKSEL_CON(57), 0, 5, MFLAGS, 8, 4, DFLAGS,
+                       PX30_CLKGATE_CON(16), 15, GFLAGS),
+
 };

 static struct rockchip_clk_branch px30_gpu_src_clk[] __initdata = {
diff --git a/include/dt-bindings/clock/px30-cru.h b/include/dt-
bindings/clock/px30-cru.h
index 644d1f5d26d0..648d7b5ae3a5 100644
--- a/include/dt-bindings/clock/px30-cru.h
+++ b/include/dt-bindings/clock/px30-cru.h
@@ -102,6 +102,7 @@
```

```
 #define SCLK_SDMMC_DIV50        87
 #define SCLK_I2S0_TX_MUX        88
 #define SCLK_I2S0_RX_MUX        89
+#define SCLK_TEST_OUT           90
```

3.2.4 RK3308

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL：

```
diff --git a/drivers/clk/rockchip/clk-rk3308.c b/drivers/clk/rockchip/clk-
rk3308.c
index 9d09c516bea5..fba208e1a09f 100644
--- a/drivers/clk/rockchip/clk-rk3308.c
+++ b/drivers/clk/rockchip/clk-rk3308.c
@@ -192,6 +192,14 @@ PNAME(mux_spdif_tx_p)              = { "clk_spdif_tx_src",
"clk_spdif_tx_frac", "mclk_i2s0_2
 PNAME(mux_spdif_rx_src_p)      = { "clk_spdif_rx_div", "clk_spdif_rx_div50" };
 PNAME(mux_spdif_rx_p)          = { "clk_spdif_rx_src", "clk_spdif_rx_frac" };
 PNAME(mux_uart_src_p)          = { "xin24m", "usb480m", "dpll", "vpll0",
"vpll1" };
+PNAME(mux_sclk_testout_p) =    { "xin24m", "armclk", "dummy", "dummy",
+                                 "aclk_peri", "hclk_peri", "clk_nandc",
+                                 "clk_sdmmc", "clk_sdio", "clk_emmc",
+                                 "clk_sfc", "dummy", "aclk_bus",
+                                 "hclk_bus", "clk_crypto", "clk_crypto_apk",
"dclk_vop",
+                                 "clk_uart0", "clk_i2c0", "clk_spi0",
+                                 "clk_tsadc", "clk_saradc", "clk_otp",
"hclk_audio"};
+
 static u32 uart_src_mux_idx[]  = { 4, 3, 0, 1, 2 };

 static struct rockchip_pll_clock rk3308_pll_clks[] __initdata = {
@@ -899,6 +907,10 @@ static struct rockchip_clk_branch rk3308_clk_branches[]
__initdata = {
        GATE(PCLK_PWM2, "pclk_pwm2", "pclk_bus", CLK_IGNORE_UNUSED,
RK3308_CLKGATE_CON(7), 13, GFLAGS),
        GATE(PCLK_CAN, "pclk_can", "pclk_bus", CLK_IGNORE_UNUSED,
RK3308_CLKGATE_CON(7), 14, GFLAGS),
        GATE(PCLK_OWIRE, "pclk_owire", "pclk_bus", CLK_IGNORE_UNUSED,
RK3308_CLKGATE_CON(7), 15, GFLAGS),
+
+       COMPOSITE(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
```

```
+               RK3308_CLKSEL_CON(73), 8, 5, MFLAGS, 0, 5, DFLAGS,
+               RK3308_CLKGATE_CON(4), 9, GFLAGS),
 };

 static struct rockchip_clk_branch rk3308_clk_ddrphy[] __initdata = {
diff --git a/include/dt-bindings/clock/rk3308-cru.h b/include/dt-
bindings/clock/rk3308-cru.h
index 5088a0f6fb02..c4707e8775b6 100644
--- a/include/dt-bindings/clock/rk3308-cru.h
+++ b/include/dt-bindings/clock/rk3308-cru.h
@@ -133,6 +133,7 @@
 #define SCLK_PWM1               120
 #define SCLK_PWM2               121
 #define SCLK_OWIRE              122
+#define SCLK_TESTOUT   123

 /* dclk */
 #define DCLK_VOP                125
```

3.2.5 RV1108

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL：

```
--- a/drivers/clk/rockchip/clk-rv1108.c
+++ b/drivers/clk/rockchip/clk-rv1108.c
@@ -159,6 +159,11 @@ PNAME(mux_dclk_hdmiphy_p)  = { "hdmiphy", "xin24m" };
 PNAME(mux_dclk_vop_p)          = { "dclk_hdmiphy", "dclk_vop_src" };
 PNAME(mux_hdmi_cec_src_p)              = { "dpll", "gpll", "xin24m" };
 PNAME(mux_cvbs_src_p)          = { "apll", "io_cvbs_clkin", "hdmiphy", "gpll"
};
+PNAME(mux_sclk_testout_p) =    { "armclk", "aclk_bus_pre", "aclk_vio0",
"aclk_vio1",
+                                 "aclk_periph", "sclk_dsp", "aclk_rkvdec",
+                                 "aclk_rkvenc", "xin24m", "dummy",
+                                 "dclk_vop", "sclk_wifi", "sclk_rga",
+                                 "sclk_isp", "aclk_vpu", "clk_venc_core"};

 static struct rockchip_pll_clock rv1108_pll_clks[] __initdata = {
        [apll] = PLL(pll_rk3399, PLL_APLL, "apll", mux_pll_p, 0,
RV1108_PLL_CON(0),
@@ -776,6 +781,10 @@ static struct rockchip_clk_branch rv1108_clk_branches[]
__initdata = {

        MMC(SCLK_EMMC_DRV,     "emmc_drv",     "sclk_emmc",  RV1108_EMMC_CON0,
1),
```

```
        MMC(SCLK_EMMC_SAMPLE,  "emmc_sample",  "sclk_emmc",  RV1108_EMMC_CON1,
1),
+
+       COMPOSITE_DIV_OFFSET(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+               RV1108_MISC_CON, 8, 4, MFLAGS, RV1108_CLKSEL_CON(39), 8, 5,
DFLAGS,
+               RV1108_CLKGATE_CON(9), 9, GFLAGS),
 };

 static const char *const rv1108_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rv1108-cru.h b/include/dt-
bindings/clock/rv1108-cru.h
index d8d0e0456dc2..b1cbd24a78eb 100644
--- a/include/dt-bindings/clock/rv1108-cru.h
+++ b/include/dt-bindings/clock/rv1108-cru.h
@@ -86,6 +86,7 @@
 #define SCLK_UART0_SRC                 127
 #define SCLK_UART1_SRC                 128
 #define SCLK_UART2_SRC                 129
+#define SCLK_TESTOUT           130

 #define DCLK_VOP_SRC                   185
 #define DCLK_HDMIPHY                   186
```

3.2.6 RV1126

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL：

```
diff --git a/drivers/clk/rockchip/clk-rv1126.c b/drivers/clk/rockchip/clk-
rv1126.c
index 05dea9fbd1f0..39905572ca13 100644
--- a/drivers/clk/rockchip/clk-rv1126.c
+++ b/drivers/clk/rockchip/clk-rv1126.c
@@ -220,6 +220,14 @@ PNAME(mux_gpll_cpll_hpll_p)                = { "gpll",
"dummy_cpll", "dummy_hpll" };
 PNAME(mux_gpll_cpll_apll_hpll_p)       = { "gpll", "dummy_cpll", "dummy_apll",
"dummy_hpll" };
 #endif

+PNAME(mux_sclk_testout_p) =    { "xin24m", "clk_rtc32k", "armclk", "dummy",
+                                 "aclk_pdbus", "hclk_pdbus", "clk_core_npu",
+                                 "aclk_pdnpu", "aclk_pdvdec",
"clk_vdec_hevc_ca",
+                                 "aclk_pdvepu", "clk_venc_core", "aclk_pdispp",
```

```
+                                "aclk_pdjpeg", "aclk_pdvi", "aclk_pdvo",
"dclk_vop",
+                                "clk_spi1", "dummy", "sclk_uart3",
+                                "aclk_pdphp", "sclk_sfc", "dummy", "dummy"};
+
 static u32 rgmii_mux_idx[]             = { 2, 3, 0, 1 };

 static struct rockchip_pll_clock rv1126_pmu_pll_clks[] __initdata = {
@@ -1378,6 +1386,10 @@ static struct rockchip_clk_branch rv1126_clk_branches[]
__initdata = {
        GATE(PCLK_DDRPHY, "pclk_ddrphy", "pclk_pdtop", CLK_IGNORE_UNUSED,
                        RV1126_CLKGATE_CON(23), 0, GFLAGS),
 #endif
+
+       COMPOSITE(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+               RV1126_CLKSEL_CON(75), 8, 5, MFLAGS, 0, 5, DFLAGS,
+               RV1126_CLKGATE_CON(23), 13, GFLAGS),
 };

 static const char *const rv1126_cru_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rv1126-cru.h b/include/dt-
bindings/clock/rv1126-cru.h
index 474bcbc546af..2f18e3127113 100644
--- a/include/dt-bindings/clock/rv1126-cru.h
+++ b/include/dt-bindings/clock/rv1126-cru.h
@@ -213,6 +213,7 @@
 #define CLK_NPUPVTM            146
 #define SCLK_DDRCLK            147
 #define CLK_OTP                        148
+#define SCLK_TESTOUT   149

 /* dclk */
 #define DCLK_DECOM             150
```

3.2.7 RK3368

1. DTS:

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTOUT>;
assigned-clock-parents = <&xin24m>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTOUT>;
clock-names = "soc_24M";
```

2. KERNEL:

```
diff --git a/drivers/clk/rockchip/clk-rk3368.c b/drivers/clk/rockchip/clk-
rk3368.c
index 009abde551aa..d5a7f720ceb2 100644
--- a/drivers/clk/rockchip/clk-rk3368.c
+++ b/drivers/clk/rockchip/clk-rk3368.c
```

```
@@ -157,6 +157,11 @@ PNAME(mux_uart3_p)         = { "uart3_src", "uart3_frac",
"xin24m" };
 PNAME(mux_uart4_p)             = { "uart4_src", "uart4_frac", "xin24m" };
 PNAME(mux_mac_p)               = { "mac_pll_src", "ext_gmac" };
 PNAME(mux_mmc_src_p)           = { "cpll", "gpll", "usbphy_480m", "xin24m" };
+PNAME(mux_sclk_testout_p) =    { "aclk_peri", "armclkb", "aclk_vio0", "dummy",
+                                 "aclk_video", "sclk_gpu_core", "dummy",
+                                 "dummy", "xin24m", "aclk_cci_pre",
+                                 "xin32k", "dummy", "dclk_vop",
+                                 "armclkl", "aclk_gpu_src", "sclk_isp"};

 static struct rockchip_pll_clock rk3368_pll_clks[] __initdata = {
        [apllb] = PLL(pll_rk3066, PLL_APLLB, "apllb", mux_pll_p, 0,
RK3368_PLL_CON(0),
@@ -881,6 +886,10 @@ static struct rockchip_clk_branch rk3368_clk_branches[]
__initdata = {
        GATE(SCLK_TIMER02, "sclk_timer02", "xin24m", CLK_IGNORE_UNUSED,
RK3368_CLKGATE_CON(24), 2, GFLAGS),
        GATE(SCLK_TIMER01, "sclk_timer01", "xin24m", CLK_IGNORE_UNUSED,
RK3368_CLKGATE_CON(24), 1, GFLAGS),
        GATE(SCLK_TIMER00, "sclk_timer00", "xin24m", CLK_IGNORE_UNUSED,
RK3368_CLKGATE_CON(24), 0, GFLAGS),
+
+       COMPOSITE_DIV_OFFSET(SCLK_TESTOUT, "sclk_testout", mux_sclk_testout_p,
CLK_SET_RATE_NO_REPARENT,
+                RK3368_MISC_CON, 0, 4, MFLAGS, RK3368_CLKSEL_CON(47), 8, 5,
DFLAGS,
+                RK3368_CLKGATE_CON(7), 1, GFLAGS),
 };

 static const char *const rk3368_critical_clocks[] __initconst = {
diff --git a/include/dt-bindings/clock/rk3368-cru.h b/include/dt-
bindings/clock/rk3368-cru.h
index 5d3531686790..c6e36b596d24 100644
--- a/include/dt-bindings/clock/rk3368-cru.h
+++ b/include/dt-bindings/clock/rk3368-cru.h
@@ -94,6 +94,7 @@
 #define SCLK_DDRCLK            139
 #define SCLK_TSP               140
 #define SCLK_HSADC_TSP         141
+#define SCLK_TESTOUT   142

 #define DCLK_VOP               190
 #define MCLK_CRYPTO            191
```

3.2.8 RK3399

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTCLKOUT1>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTCLKOUT1>;
clock-names = "soc_24M";
```

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru SCLK_TESTCLKOUT2>;
assigned-clock-rates = <24000000>;
clocks = <&cru SCLK_TESTCLKOUT2>;
clock-names = "soc_24M";
```

3.2.9 RK3566/8

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&pmucru CLK_WIFI>;
assigned-clock-rates = <24000000>;
clocks = <&pmucru CLK_WIFI>;
clock-names = "soc_24M";
```

3.2.10 RK3588

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru CLK_CIFOUT_OUT>;
assigned-clock-rates = <24000000>;
clocks = <&cru CLK_CIFOUT_OUT>;
clock-names = "soc_24M";
```

3.2.11 RK3528

- CLK_TESTOUT: 暂不开发对外使用，用于内部自测信号使用。
- CLK_REFOUT: 时钟频率默认只有24M，可开关，不支持频率修改。
- CLK_DEEPSLOW: 时钟频率默认32K，不支持开关和频率修改。

所以只需配置Pinctrl即可。

3.2.12 RK3576

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&test_clkout>;
assigned-clocks = <&cru CLK_MIPI_CAMERAOUT_M0>;
assigned-clock-rates = <24000000>;
clocks = <&cru CLK_MIPI_CAMERAOUT_M0>;
clock-names = "soc_24M";
```

3.2.13 RK3538/9

1. DTS：

```
pinctrl-names = "default";
pinctrl-0 = <&ref_clkm0_pins>;
assigned-clocks = <&cru CLK_REFOUT>;
assigned-clock-rates = <24000000>;
clocks = <&cru CLK_REFOUT>;
clock-names = "refout";
```

备注：

dts中pinctrl的控制，如果疑问参考pinctrl的文档。
RK3568\RK3588 IO比较丰富，如果想要其他的IO输出CLK，请先查看IO和CLK的对应关系，然后按照示例操作。